TRICOFF Coffee Kiosk: Prototype Build Blueprints

Rev. 2, airpot version: no pumps, no tubes, fully offline. Everything needed to buy, wire, build and test the machine.

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What changed from Rev. 1 (pump version)

- **Pumps, tubes, steel straws and MOSFETs are gone.** Coffee stays inside 3 ordinary push-top airpots; a metal-gear hobby servo (MG996R) presses each airpot's top.
- **Nothing that touches coffee is modified,** so no food-grade or 80 °C-rated parts have to be hunted down online.
- **Almost every part comes from ordinary stores:** mall kitchenware and hardware shops. Only the electronics (ESP32, 3 servos, 2 buck converters, capacitor, cables) need an electronics shop or Shopee/Lazada, and all of them come from the same place.
- **Fully offline, no router:** the ESP32 itself broadcasts the kiosk's own Wi-Fi network (TRICOFF). No internet, no router and no phone hotspot are needed.
- **Pour size is now 200 ml** so an 8 oz (237 ml) cup does not overflow.

1. How the machine works

Step	What happens
1	The customer picks a coffee on the kiosk screen (the TRICOFF website running on the laptop, opened at **http://localhost:3000/kiosk**).
2	The screen shows a QR code and which slot to put the cup in (1, 2 or 3).
3	The customer scans the QR with their phone. The phone opens the simulated payment page from the laptop over the kiosk Wi-Fi (**http://192.168.4.200:3000/pay/…**). No real money and no internet are involved.
4	On **Pay**, the laptop tells the ESP32 to pour that order. On **Fail**, nothing pours.
5	The ESP32 makes that airpot's servo press its push-top the calibrated number of times (≈200 ml), then reports back and the screen shows “Done”.

Main design decisions

- **Hot coffee without a heater.** Coffee is brewed beforehand and kept hot in 3 vacuum airpots (stays hot for several hours).
- **No pumps, no tubes.** The airpot's own push-top does the pouring. A metal-gear servo swings a short arm down onto it; each press gives about the same amount, so the ESP32 counts presses.
- **Three separate airpots.** Regular, Sweetened and Decaf never share a part, so Decaf stays decaf.
- **Electronics high on the back wall,** above the airpot tops and away from the spouts, so a spill cannot reach the boards.
- **No 220 V inside the machine.** A 12 V wall adapter powers everything. Unplugging it stops the machine instantly.
- **Works with no internet and no router.** The ESP32 broadcasts the TRICOFF Wi-Fi; the laptop and the customer's phone join it.
- **Uses what you already have:** laptop (server + screen), phone (customer side), micro-USB cable, basic tools.

Safety rules

- The coffee is about 80 °C. Use hot-drink cups and keep hands away from the spouts while pouring.
- Emergency stop = pull the 12 V adapter plug.
- Keep the kiosk lid closed and its two front screws in while the machine is powered: the servo arms move without warning.
- Test with water until step 15 of the assembly. Use real coffee only after the water test passes.

2. Parts list

One choice per part. Costs are estimates in Philippine pesos; replace them with the prices you actually pay. The photos in 2.1 show what each part looks like.

#	What to buy	Qty	Where to buy	Est. ₱
1	ESP32 DevKit V1 (ESP-WROOM-32), 38-pin	1	Electronics shop / online	300–450
2	MG996R metal-gear servo (with horns and screws)	4 (3 + 1 spare)	Electronics shop / online	600–1,200
3	LM2596 buck converter module (adjustable), for the ESP32	1	Electronics shop / online	60–120
4	XL4015 buck converter, 5 A, for the servos	1	Electronics shop / online	120–200
5	Electrolytic capacitor, 1000 µF 25 V	1	Electronics shop	10–30
6	Servo extension cable, 3-wire, 50 cm	3	Electronics shop / online	60–120
7	Push-top airpot, stainless, 2.2–2.5 L	3	SM / hypermarket kitchenware	1,500–3,600
8	12 V 5 A adapter, round DC plug 5.5 × 2.1 mm	1	Electronics / CCTV shop	300–450
9	DC barrel jack (female) with screw terminals, 5.5 × 2.1 mm	1	Electronics / CCTV shop	20–40
10	Inline blade fuse holder + 5 A blade fuse	1	Auto supply / hardware	50–100
11	18 AWG stranded wire, red + black, 3 m each	1 set	Hardware / auto supply	100–200
12	Female-to-female jumper wires, 40 pcs, 20 cm	1	Electronics shop / online	60–100
13	12-way screw terminal strip (barrier type)	1	Hardware / electrical	40–80
14	Heat-shrink tubing assortment	1	Electronics / auto supply	50–100
15	¼" plywood, 4 × 8 ft sheet	1	Hardware / lumber	500–900
16	1 × 1" wood batten, 8 ft	3	Hardware / lumber	120–240
17	Wood screws #6 × ¾" (100 pcs) + #4 × ⅜" (20 pcs)	1 set	Hardware	80–150
18	M4 × 20 mm bolts with nuts and washers	10 sets	Hardware	40–80
19	Wood glue, 250 ml	1	Hardware	80–150
20	2" steel butt hinges	1 pair	Hardware	60–120
21	Clear varnish, small can	1	Hardware	150–250
22	8 oz paper cups for hot drinks, 50 pcs	1	Supermarket / packaging shop	120–200
23	Zip ties, 100 pcs	1	Hardware	30–60
	TOTAL (estimate)			**4,450–8,940**

Already have (₱0)

- Laptop: runs the TRICOFF server and shows the kiosk screen.
- Phone: joins the TRICOFF Wi-Fi, scans the QR and taps Pay/Fail (as the customer).
- Micro-USB **data** cable for uploading code to the ESP32 (some cables only charge).
- 3 plastic bottle caps (press pads) and a plastic drinking straw (cut into short spacers).
- Tools: multimeter, soldering iron, drill with 4, 6 and 12 mm bits, hand saw, coping saw or utility knife and a flat file (servo cut-outs), screwdrivers, clamps or heavy books, heat gun or lighter, measuring cup.

2.1 Parts reference photos

One box per part, numbered as in the parts list. Photos are added from the actual listings or store items, so they match what you buy.



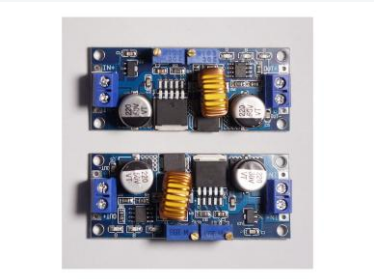
1. ESP32 DevKit V1 (ESP-WROOM-32), 38-pin
Look for: Silver metal can marked ESP-WROOM-32, 19 pins per side, micro-USB port, EN and BOOT buttons.



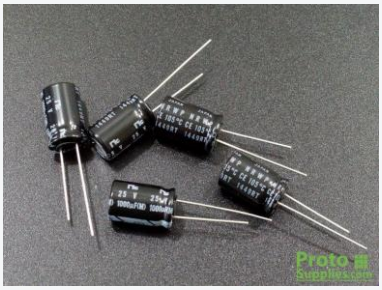
2. MG996R metal-gear servo (with horns and screws)
Look for: Black servo labelled MG996R, metal gears, 3-wire plug (brown, red, orange); comes with plastic horns and screws. Not SG90/MG90 (too weak).



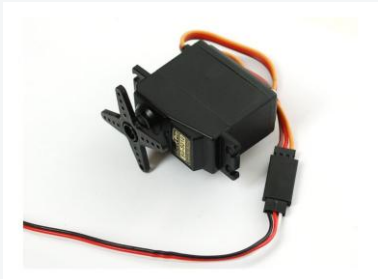
3. LM2596 buck converter module (adjustable), for the ESP32
Look for: Small blue board, large black coil, blue trimmer with a brass screw, IN+ IN− OUT+ OUT− pads.



4. XL4015 buck converter, 5 A, for the servos
Look for: Board with a heatsink and two blue trimmers (voltage and current), marked IN+ IN− OUT+ OUT−.



5. Electrolytic capacitor, 1000 µF 25 V
Look for: Round can marked 1000µF 25V; the stripe marks the minus leg.



6. Servo extension cable, 3-wire, 50 cm
Look for: Brown/red/orange cable with a plug at each end, made for hobby servos.



7. Push-top airpot, stainless, 2.2–2.5 L
Look for: Big push button on the lid, lock lever, spout at the front, handle at the back. Check the sizes in section 7 before buying.



8. 12 V 5 A adapter, round DC plug 5.5 × 2.1 mm
Look for: Label says 12V 5A (or more); round barrel plug.



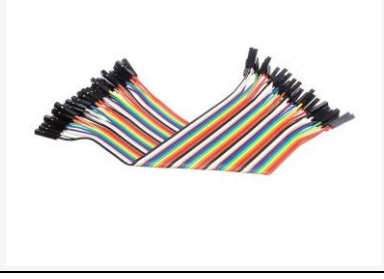
9. DC barrel jack (female) with screw terminals, 5.5 × 2.1 mm
Look for: Female barrel socket with two screw terminals marked + and −.



10. Inline blade fuse holder + 5 A blade fuse
Look for: Holder for standard (ATC) car blade fuses; tan 5 A fuse.



11. 18 AWG stranded wire, red + black, 3 m each
Look for: Flexible multi-strand wire (“automotive wire”), red and black.



12. Female-to-female jumper wires, 40 pcs, 20 cm
Look for: Rainbow ribbon of wires with square sockets at both ends.

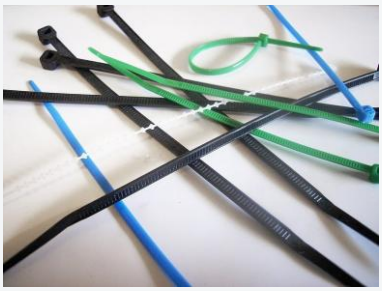
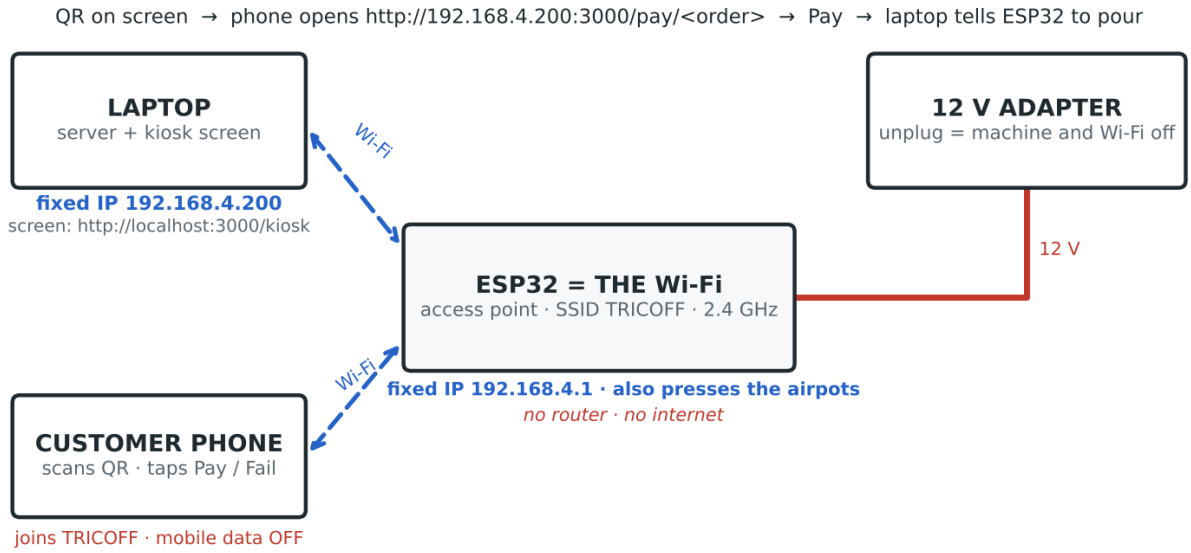
		
13. 12-way screw terminal strip (barrier type) Look for: Black strip with 12 positions and screws on both sides.	**14. Heat-shrink tubing assortment** Look for: Box of assorted sizes, 2–10 mm.	**15. ¼" plywood, 4 × 8 ft sheet** Look for: Flat sheet, no warping or cracked edges.
		
16. 1 × 1" wood batten, 8 ft Look for: Straight pieces; sight along each one before buying.	**17. Wood screws #6 × ¾" (100 pcs) + #4 × ⅜" (20 pcs)** Look for: Flat-head wood screws.	**18. M4 × 20 mm bolts with nuts and washers** Look for: Machine bolts, nuts and flat washers (for the lid hinges).
		
19. Wood glue, 250 ml Look for: White PVA wood glue.	**20. 2" steel butt hinges** Look for: Plain steel hinges with screw holes.	**21. Clear varnish, small can** Look for: Clear polyurethane; seals the cup shelf.
		
22. 8 oz paper cups for hot drinks, 50 pcs Look for: Hot-drink cups, no taller than 9.5 cm.	**23. Zip ties, 100 pcs** Look for: Assorted 100 mm and 200 mm.	

Photo sources (for identification only; brands shown are examples, not requirements; some photos cropped): 1 domoticx.net; 2 Tower Pro; 3 Matthew Berardi, Wikimedia Commons, CC BY-SA 4.0; 4 Wikimedia Commons, CC0; 5 ProtoSupplies; 6, 8, 9 Adafruit; 7 Hospitality Superstore; 10 Del City; 11 Pico Wiring via Altex; 12 oomlout, Wikimedia Commons, CC BY-SA 2.0; 13 Philmore via Vetco; 14 Dmitry G, Wikimedia Commons, CC BY-SA 3.0; 15 Downtowngal, Wikimedia Commons, CC BY-SA 4.0; 16 Alexandria Moulding; 17 Dmitry Makeev, Wikimedia Commons, CC BY 3.0; 18 andersen_mrjh, Wikimedia Commons, CC BY-SA 2.0; 19 Expo Tools; 20 Ronak45, Wikimedia Commons, CC0; 21 Minwax; 22 Pactiv Evergreen; 23 Silverxxx, Wikimedia Commons, CC BY-SA 3.0.

3. Offline network and software

The kiosk never needs the internet or a router. The ESP32 runs in **access-point mode**: it broadcasts its own Wi-Fi network called TRICOFF, and the laptop and the customer's phone join it like any Wi-Fi. “localhost” only works on the laptop itself, so the phone reaches the laptop by its fixed IP address.



Blue dashed = Wi-Fi from the ESP32. Only the kiosk's own devices are on this network; nothing goes to the internet.

3.1 Settings

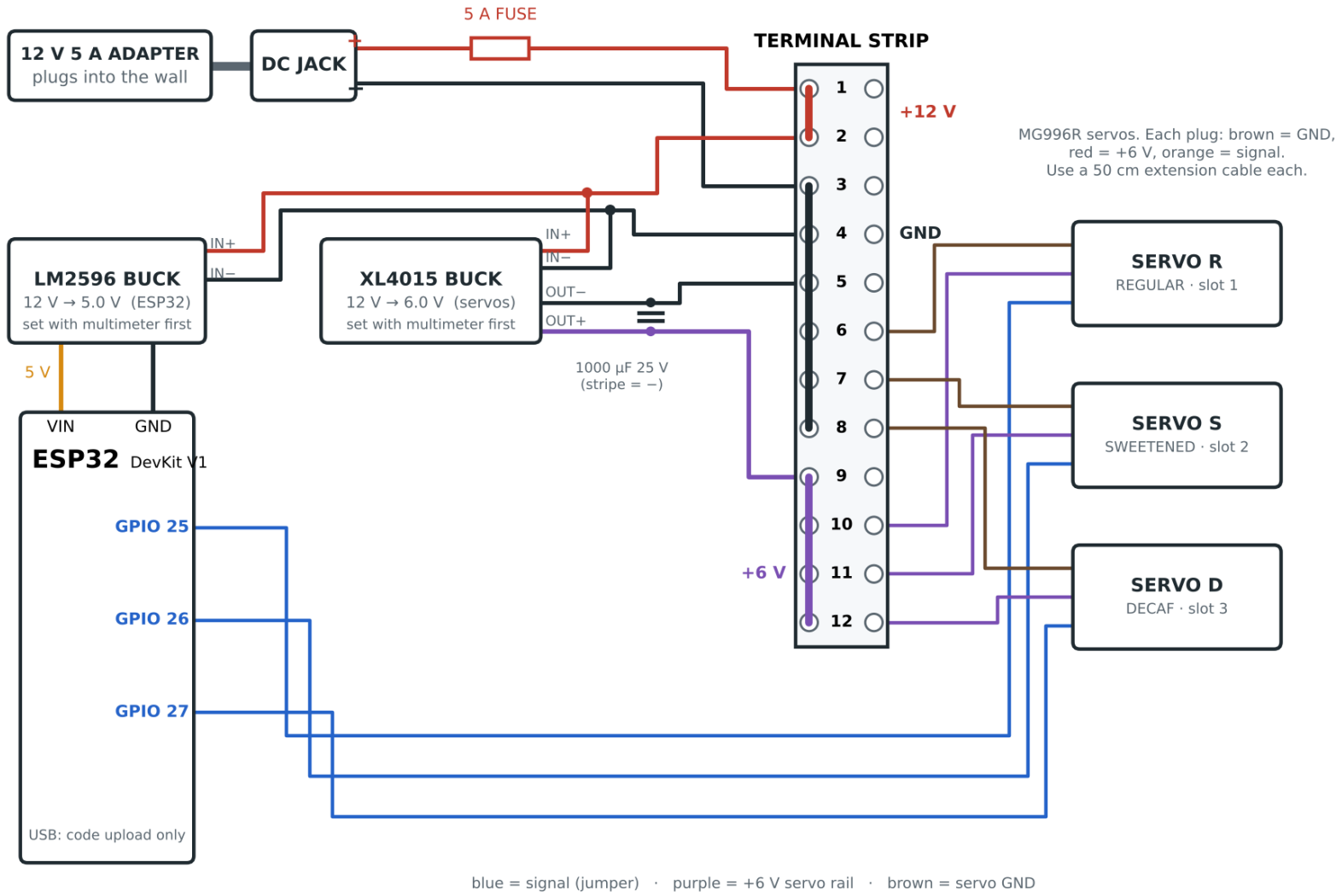
Item	Setting	Why
Wi-Fi network	Made by the ESP32 in access-point mode: name **TRICOFF**, WPA2 password of 8+ characters, 2.4 GHz. Allow up to **10 devices** in the code (the default is 4).	No router needed. The network is on whenever the kiosk is plugged in.
Laptop IP	**192.168.4.200**, fixed. In Windows, open the TRICOFF Wi-Fi properties → IP assignment → Manual: IP 192.168.4.200, subnet mask 255.255.255.0, gateway blank.	The QR code points to this address. .200 stays clear of the addresses the ESP32 hands to phones (.2 upward).
ESP32 IP	**192.168.4.1** (the ESP32's own address in access-point mode).	The laptop sends pour commands to this address.
Server	Listens on **0.0.0.0, port 3000** (not only 127.0.0.1).	So the phone can reach it, not just the laptop.
Kiosk screen	Laptop browser, full screen: **http://localhost:3000/kiosk**	localhost is fine for the laptop's own screen.
QR code	Encodes **http://192.168.4.200:3000/pay/<orderId>**	Opens on the phone over the TRICOFF Wi-Fi.
Windows firewall	When Windows asks, allow the server on **Private** networks; set the TRICOFF Wi-Fi to Private.	Otherwise the phone cannot open the page.
Website files	Fonts, CSS, JavaScript and the QR library are saved inside the project. **No CDN or Google Fonts links.** Make the QR on the laptop (e.g. the npm “qrcode” package).	Anything loaded from the internet breaks offline.
Customer phone	Join TRICOFF Wi-Fi; if it says “no internet”, choose to stay connected; turn mobile data **off**.	Stops the phone from routing around the kiosk Wi-Fi.
Laptop internet	While the laptop is on TRICOFF it has no internet. That is normal; download everything the project needs beforehand.	The demo runs fully offline.
If the ESP32 restarts	The TRICOFF Wi-Fi disappears for a few seconds. Tick “Connect automatically” on the laptop so it rejoins by itself.	The kiosk screen shows “Machine offline” until it is back.

3.2 Messages between the parts

From → to	Message	What it does
Kiosk screen → server	POST /api/orders {drink}	Creates an order ID and returns the QR link and the cup slot.

From → to	Message	What it does
Phone → server	GET /pay/<orderId>, then POST /api/orders/<id>/pay or /fail	Simulated payment page with Pay and Fail buttons.
Server → ESP32	POST http://192.168.4.1/pour {order, line, presses}	Starts the pour. **The ESP32 ignores an order ID it has already poured,** so a retry or double tap never pours twice.
ESP32 → server	POST /api/orders/<id>/done	Screen shows “Done – take your coffee”.
Server → ESP32	GET http://192.168.4.1/status every few seconds	Kiosk screen shows “Machine offline” and blocks new orders if the ESP32 stops answering.

4. Wiring diagram



Red = +12 V · Black = GND · Orange = 5 V · Purple = +6 V servo rail · Brown = servo GND · Blue = signal (jumper wire). Dots are connections; wires that cross without a dot are NOT connected.

4.1 Terminal strip

Bridge positions 1–2 together on the left screws (short pieces of red 18 AWG), 3–8 together (black) and 9–12 together (red, marked “6 V” with tape). Then connect each item to its own position on the right screws. Label the strip with a marker.

Pos.	Rail	Right-side wire goes to
1	+12 V	From the 5 A fuse (which comes from DC jack +)
2	+12 V	LM2596 IN+ and XL4015 IN+ (two wires under one screw)
3	GND	From DC jack −
4	GND	LM2596 IN− and XL4015 IN−
5	GND	XL4015 OUT− and capacitor − leg (stripe)
6	GND	Servo R (Regular) brown wire
7	GND	Servo S (Sweetened) brown wire
8	GND	Servo D (Decaf) brown wire
9	+6 V	XL4015 OUT+ and capacitor + leg
10	+6 V	Servo R red wire
11	+6 V	Servo S red wire
12	+6 V	Servo D red wire

4.2 ESP32 connections

From	Connects to	Wire
LM2596 OUT+ (5.0 V)	ESP32 **VIN**	Jumper, orange if you have it
LM2596 OUT−	ESP32 **GND**	Jumper, black
GPIO 25	Servo R (Regular) orange signal wire	Jumper
GPIO 26	Servo S (Sweetened) orange signal wire	Jumper
GPIO 27	Servo D (Decaf) orange signal wire	Jumper
ESP32 USB	Laptop, only when uploading code (unplug the 12 V adapter first)	Micro-USB data cable

4.3 Servo cables

Each MG996R has a 3-pin plug. Plug a 50 cm extension cable into each servo, then at the free end cut off the plug and strip the three wires.

Servo wire	Means	Goes to
Brown	GND	Strip position 6 (R), 7 (S) or 8 (D)
Red	+6 V	Strip position 10 (R), 11 (S) or 12 (D)
Orange	Signal	Solder to half of a female jumper wire, cover with heat-shrink, push onto GPIO 25 (R), 26 (S) or 27 (D)

The ESP32 and the servos share ground through the two buck converters, so no extra ground wire is needed. Check it: the multimeter beeps between the ESP32 GND pin and strip position 3.

4.4 Pour settings (for the code)

Setting	Value
Library	ESP32Servo (Arduino IDE library manager). One servo object per GPIO.
REST angle	Arm level, bottle cap 1 cm above the push-top. Start at **90°** and adjust in step 14.
PRESS angle	Arm about **50° further down**. Start at **140°**; if the arm moves up instead, use **40°**. All three servos face the same way, so one value fits all.
One press	Go to PRESS → hold **0.4 s** → back to REST → wait **0.6 s**

Setting	Value
Presses per cup	round(200 ÷ ml per press), measured in step 15 for each airpot. Example: 25 ml per press → 8 presses.
At start-up	Send every servo to REST first thing in setup(), so the arms start up and away from the push-tops.
Never	Leave a servo at PRESS for more than 1 s. Pushing against a stop makes it draw full current and heat up.

4.5 Wiring rules

- Set the LM2596 to 5.0 V and the XL4015 to 6.0 V with a multimeter **before** connecting anything to them. Never set the servo supply above 6.8 V.
- Never power the servos from the ESP32's 5V or 3V3 pins. They run only from the 6 V rail.
- GPIO 25/26/27 were chosen because they stay quiet while the ESP32 boots, so no servo jumps by itself at power-up.
- Where a jumper wire meets a screw terminal, cut off the plug, strip 6 mm, twist and screw it in.
- Leave a 30 cm slack loop in each servo cable and zip-tie it along board H, so the lid can open.
- Cover every solder joint with heat-shrink. Pull each wire gently after tightening; nothing should come loose.

5. Enclosure

A plywood box, **64 × 40 × 60 cm** (W × D × H), open at the front between 18 and 42 cm. The three airpots stand on the floor at the back with their spouts over the cup shelf. Each servo sits on a small plate hanging from a board under the lid, with a short arm reaching over its airpot's push-top; the lid lifts on two back hinges to refill the airpots. The electronics sit on the back wall, 42–57 cm up. All 6 mm plywood joints are screwed into 1 × 1" battens, never into the thin plywood edge.

5.1 Plywood cut list (all sizes in cm)

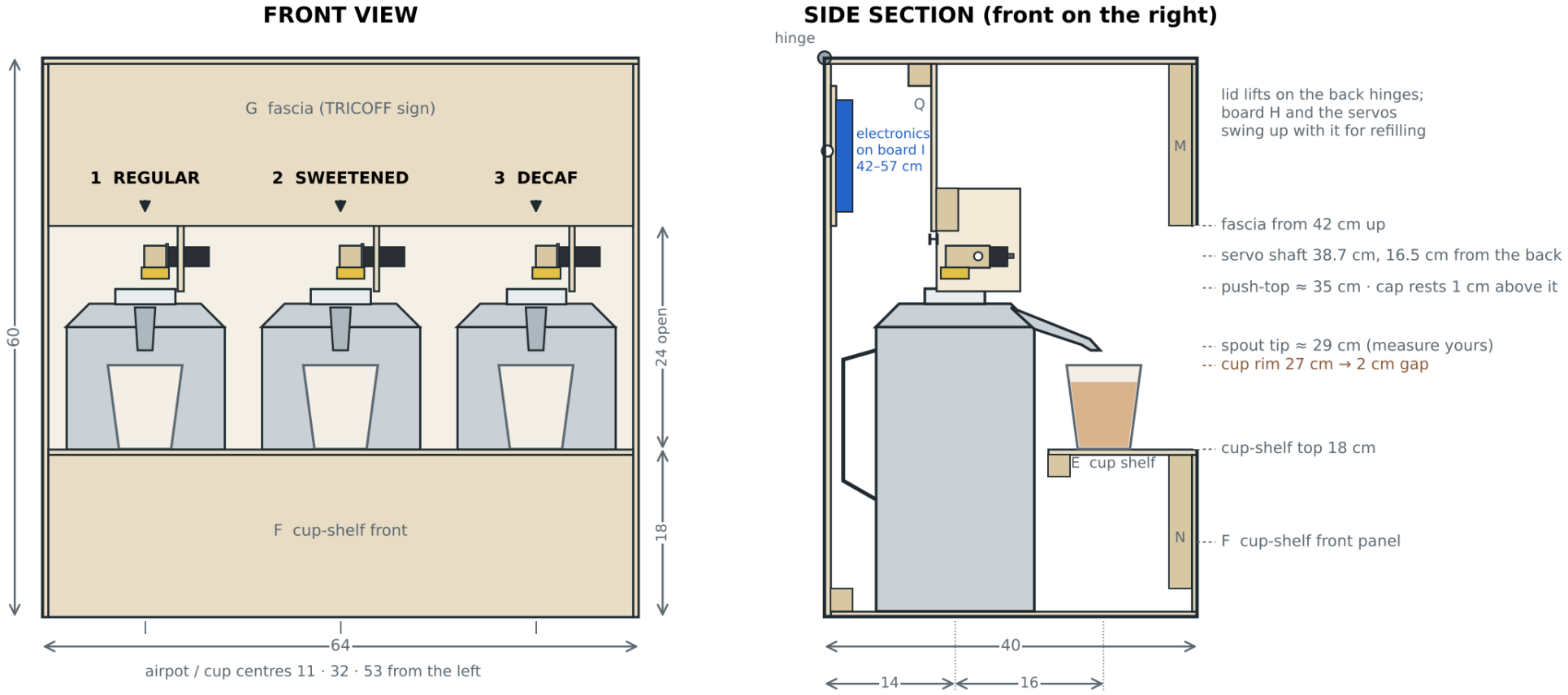
Ref	Piece	Qty	Size (W × H)	Holes / notes
A	Side panel	2	40 × 59.4	—
B	Back panel	1	62.8 × 59.4	12 mm hole for the adapter cable: 50 cm up, 5 cm from the right edge (seen from behind). Hinge bolt holes drilled at step 11.
C	Lid	1	64 × 40	Hinged at the back (M4 bolts). Two #6 screws at the front corners, 1.8 cm in from the side and front edges, into battens M.
D	Bottom	1	62.8 × 39.4	—
E	Cup shelf	1	62.8 × 16	Top at 18 cm (see section 7). Varnish. Mark 3 cup circles, 8 cm across, directly under the spout tips (drawn at 10.4 / 31.4 / 52.4 cm from E's left edge, 10 cm from its front edge).
F	Cup-shelf front	1	62.8 × 16.8	Closes the space under the shelf. Varnish the front.
G	Fascia	1	62.8 × 17.4	From 42 cm up. Write 1 REGULAR · 2 SWEETENED · 3 DECAF above slots at 11 / 32 / 53 cm from the box's left edge.
H	Servo board	1	61.8 × 18	Hangs from the lid and carries the 3 servo plates. Its front face is 12 cm from the lid's back edge.
S	Servo plate	3	9 × 11 (D × H)	Stands front-to-back under H, back edge against H's front face, bottom at 35 cm. Inner face 14.5 / 35.5 / 56.5 cm from the box's left edge. Cut-out 41 × 20 mm, its corner 3.5 cm from the plate's back edge and 2.7 cm up from its bottom edge (check with your servo).
I	Electronics board	1	40 × 15	Glued to the back panel inside, centred, 42–57 cm up. Boards are screwed to it with #4 × ⅜".

5.2 Battens (1 × 1" wood, cut from 3 pieces of 8 ft)

Ref	Use	Qty	Length
K	Vertical, in both back corners (side to back), from the bottom panel up to the top	2	58.8 cm
L	Bottom cleats along both side panels (bottom panel screws up into them)	2	36.4 cm
R	Back bottom rail between the K battens (joins back and bottom)	1	58.0 cm
N	Front verticals behind panel F, 3.0–17.4 cm up	2	14.4 cm
P	Cup-shelf cleats on both side panels, under shelf E	2	13.0 cm
T	Stiffener under the back edge of shelf E	1	58.0 cm
M	Front verticals behind fascia G, 42–59.4 cm up (lid front screws go into their tops)	2	17.4 cm
Q	Hanger rail glued to the top of board H; the lid is screwed down into it	1	58.0 cm
J	Stop blocks on the floor that put each airpot back in the same spot	6	5.0 cm
V	Cleats joining each servo plate S to board H (glued and screwed to both)	3	4.6 cm
W	Press arms screwed to the servo horns; a bottle cap is glued under one end	3	4.7 cm

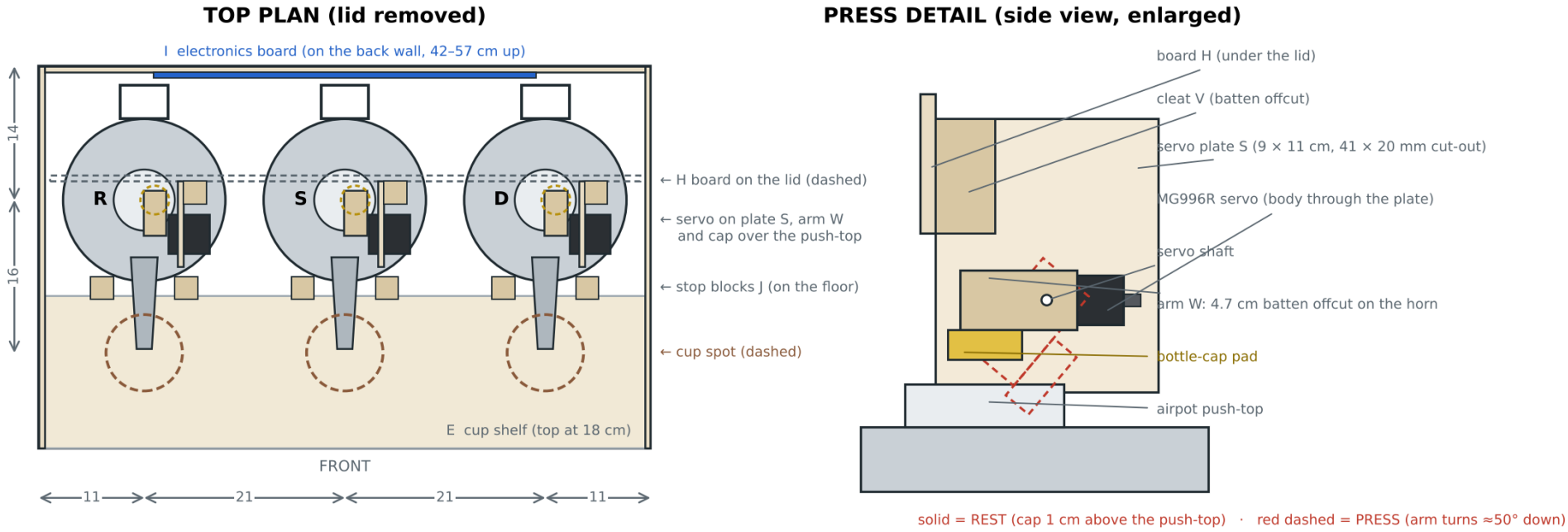
Total ≈ 5.12 m of the 7.3 m bought (cutting plan in 5.5). Glue and screw battens to the side panels first, then screw every other panel into the battens.

5.3 Front view and side section



Scale drawing, dimensions in cm. Airpots drawn at a typical 2.5 L size; measure yours (section 7) before cutting.

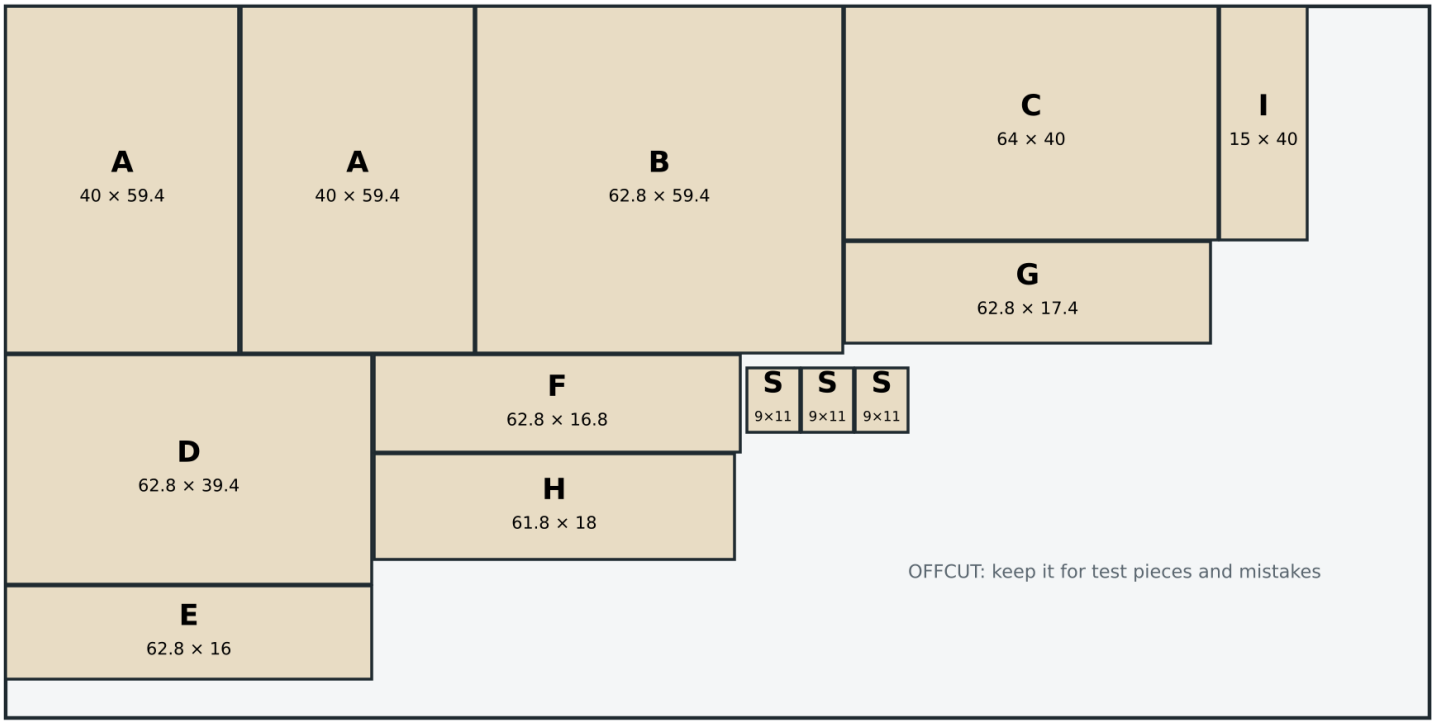
5.4 Top plan and press detail



Each servo sits beside its airpot with its arm reaching over the push-top. The ESP32 swings the arm down and back up a set number of times per cup.

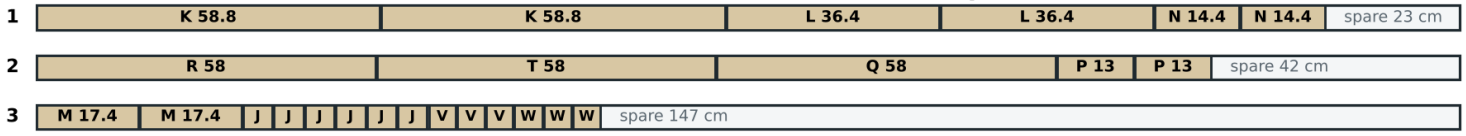
5.5 Plywood and batten cutting layout

¼" PLYWOOD SHEET · 244 × 122 cm (8 × 4 ft)



All sizes in cm, width × height. Ask the hardware store to cut on their panel saw; allow ~3 mm per saw cut.

1 × 1" BATTENS · three 8 ft (243.8 cm) pieces



Letters match the cut lists in 5.1 and 5.2. Everything fits on one 4 × 8 ft sheet and three 8 ft battens.

6. Assembly and test steps

Do the steps in order. Do not move on until the pass check is met. Use water until step 15.

Step	Stage	What to do	Pass check
1	Bench	Set the LM2596 to 5.0 V and the XL4015 to 6.0 V with the multimeter, nothing connected to their outputs.	Reads 4.95–5.10 V and 5.9–6.1 V.
2	Bench	Prepare the terminal strip: bridge 1–2 (+12 V), 3–8 (GND) and 9–12 (+6 V). Label it.	Beeps within each group; no beep between 1, 3 and 9.
3	Bench	Wire DC jack + → fuse → position 1, and jack − → position 3. Plug in the adapter.	12 V between positions 1 and 3. Unplug again.
4	Bench	Wire both buck converters, the capacitor and the ESP32 on a table, following section 4. Leave the servos unplugged.	With 12 V on: 6 V between positions 9 and 3; 5 V at the ESP32 VIN pin.
5	Bench	Upload the servo test code over USB (12 V unplugged while USB is in). Unplug USB, plug in 12 V.	The ESP32 starts by itself from the LM2596.
6	Bench	Plug the 3 servos (no arms yet) into their extension cables and the strip. Command REST and PRESS on each.	The right servo turns for each command; nothing twitches at power-up; the ESP32 never resets.
7	Network	Upload the Wi-Fi + pour code. Connect the laptop to TRICOFF and give it the fixed IP 192.168.4.200 (section 3). Start the server. Connect the phone to TRICOFF.	Phone opens http://192.168.4.200:3000; the kiosk screen shows “Machine online”.
8	Box	Have the plywood cut to 5.1 and cut the battens (5.2). Sand all edges.	All 13 plywood pieces and the battens ready and labelled.
9	Box	Drill the 12 mm cable hole in B. Varnish the top and front edge of E and the front of F. Let it dry 24 h.	Dry, not sticky.
10	Box	Assemble: battens onto sides (K, L, N, P, M) → back B + rail R → bottom D → front F → shelf E with T → fascia G.	Box is square and does not wobble.
11	Box	Lid: glue and screw Q to the top of H. Screw lid C down into Q (H centred, front face 12 cm from C's back edge). Bolt the hinges to C and B.	Lid opens past upright and closes flat; H touches nothing on the way.
12	Fit-out	Glue board I to the back wall 42–57 cm up. Mount the strip, both buck converters and the ESP32 on it. Feed the adapter cable through the back hole and zip-tie it inside.	No board is below 42 cm; the cable cannot be pulled out.
13	Fit-out	Stand the airpots: handle against the back wall, centres 11 / 32 / 53 cm from the left. Glue two stop blocks J in front of each base.	Each airpot goes back to the same spot every time.
14	Fit-out	Screw each servo into its plate S. Glue and screw cleat V to the plate, then screw the plate to the front of H (screws through H into V). Screw arm W onto each horn and glue a bottle cap under its free end. With the lid closed and the servo at REST, each cap sits 1 cm above its push-top. Route the cables with a slack loop. Put in the 2 lid screws.	Lid opens fully without pulling any cable; each cap is over its push-top.
15	Test	Water test and calibration: fill the airpots with ~85 °C water, lock lever on OPEN. Adjust each PRESS angle until a press gives a steady squirt. Run 10 presses per line into the measuring cup, 3 times. Work out ml per press and set presses per cup for 200 ml.	200 ml ±10 ml, 5 times in a row per line; nothing splashes outside the cup; the servos do not buzz at REST.
16	Test	With the laptop and phone on TRICOFF only, run the full flow for each coffee: choose → QR → Pay → pour. Then try Fail.	The right airpot pours for each choice; Fail never pours.
17	Test	While idle, unplug the 12 V adapter for 10 seconds and plug it back in. Then tap Pay twice quickly on one order.	Laptop and phone rejoin TRICOFF by themselves; screen shows “Machine online” again; nothing pours by itself; the order pours only once.
18	Final	Real coffee test, then rinse the airpots.	Each slot gives the right coffee; Decaf tastes of decaf only.

7. Check these when the parts arrive

The drawings assume typical part sizes. Measure these before cutting any wood.

Check	Needs to be	If it is not
Airpot width	Max 18 cm across the body (side to side, without the handle)	Make the box wider by 3 × the difference and space the centres to match.
Airpot height	Max 40 cm to the top of the push-top (drawings use 35 cm)	Up to 40 cm: mount the servo plates higher on board H. Taller: make the sides, back and board H taller by the difference.
Airpot depth	Back of handle to front of body max 22 cm	Cut shelf E narrower by the difference (from its back edge) and shorten cleats P to match, so the body clears the shelf (its back edge is 24 cm from the back).
Spout tip height	27–32 cm above the table	Cup-shelf top = spout tip − 11 cm. Drawings use 29 → 18 cm. If different, move cleats P, cut F to (shelf top − 1.2) and N to (shelf top − 3.6).
Spout reach	Spout tip at least 27 cm in front of the back of the handle	Mark each cup circle directly under the real spout tip; the whole cup base must still stand on the shelf.
Servo type	MG996R with metal gears (the box or label says metal gear)	Exchange it; plastic-gear SG90/MG90 servos are too weak.
Servo strength	With arm W and the cap fitted, hold the servo over a water-filled airpot and command PRESS: water comes out	Move the cap closer to the shaft (a shorter arm presses harder) or raise the XL4015 to 6.5 V (max 6.8 V).
Cup height	Max 9.5 cm	Lower the cup shelf so the spout stays about 2 cm above the cup.
Servo cut-out	The servo slides into its 41 × 20 mm cut-out and its flanges rest on the plate	File the cut-out bigger. Measure your servo first; sizes vary by a millimetre or two.
Adapter plug	Fits the DC jack (both 5.5 × 2.1 mm)	Exchange the jack for the matching size.

8. Daily use and cleaning

Before the demo

- Preheat: fill each airpot with boiling water, wait 5 minutes, empty it.
- Brew the coffee. Mix the sugar into the Sweetened batch. Fill each airpot while the coffee is still very hot (85 °C or more), up to the max line.
- Close each airpot's lid, set its lock lever to OPEN, and stand it against its stop blocks.
- Close the kiosk lid and put in the two front screws.
- Plug in the 12 V adapter and wait for the TRICOFF Wi-Fi to appear. Connect the laptop to it, start the server, and check the kiosk screen says “Machine online”.
- Prime: pour one test cup per line into a spare cup.

After use

- Unplug the 12 V adapter. Remove the two lid screws and open the lid.
- Empty the airpots, rinse with hot water, and press each push-top by hand several times to flush the spout.
- Wipe the cup shelf. Leave the airpot lids open to dry.
- Once a week: clean the airpots as their manual says (warm water and a little food-grade cleaner or baking soda), then rinse twice with plain water.